Run LLMs locally on your Ubuntu machine with integrated AMD-GPU

Why run LLMs locally?

- Pros:
  - Privacy
  - Security
  - Cost control
  - Control over models used
  - Consistent quality of answers
- Cons:
  - Not as scalable as the cloud providers
  - Not suitable for big models

Privacy in the cloud?

Why OpenAI chose ClickHouse for petabyte-scale observability

ClickHouse Team
Jun 30, 2025 - 8 minutes read

Think your observability numbers are scary? Try walking a mile in OpenAI's shoes.

Every day, the company ingests petabytes of log data—the equivalent of 500 Libraries of Congress or 2 billion iPhone photos. If you snapped a photo every second, it would take 60 years to match what OpenAI logs in a single day. You would need a pallet of hard drives just to store it. And that volume is growing by more than 20% each month.

These features have already been validated in large-scale observability deployments. Companies such as Tesla, Anthropic, OpenAI, and Character.AI rely on ClickHouse to analyze observability data at petabyte scale, demonstrating its ability to meet the demands of modern workloads.

ChatGPT Privacy Leak 2025: Deep Dive, Real-World Impact, and Industry Lessons

Ismail Kovvuru Follow 7 min read · Aug 2, 2025

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The recent revelation that **thousands of ChatGPT conversations became accessible via Google search** in 2025 has changed conversations in the tech world about AI, user trust, and product design.

ShadowLeak: ChatGPT revealed personal data from emails to attackers

Using a combination of different manipulation techniques, the OpenAI-LLM was tricked into leaking private data. What did Sam Altman know about it?

leaked-system-prompts

Description

This repository is a collection of leaked system prompts from widely used LLM based services.

Requirements

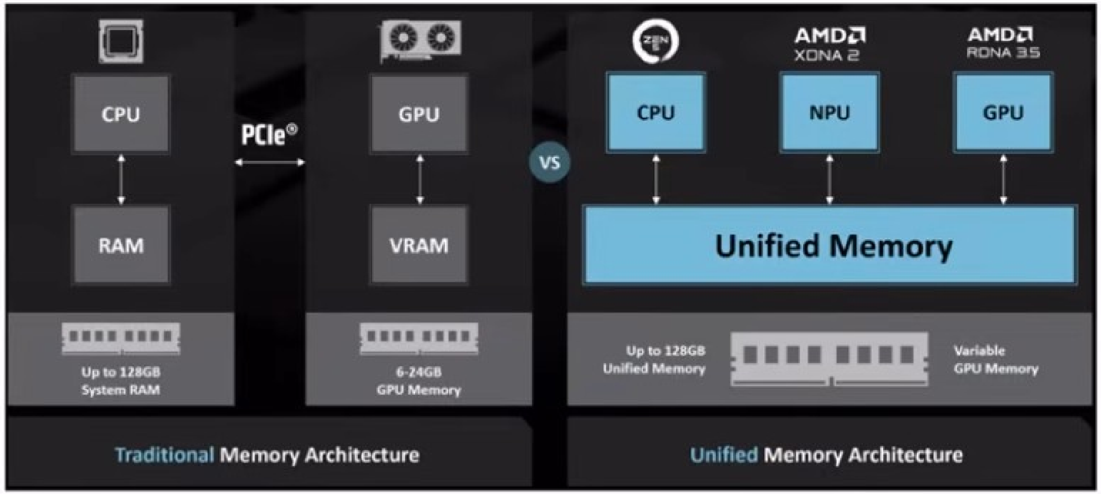
- Integrated AMD GPU
  - supported by Kernel builtin **Vulkan** API driver (RADV)
  - Ryzen 7 PRO 7840U (10 TOPs)
  - Ryzen AI 9 HX 370 (50 TOPs)
  - Ryzen AI Max+ 395 (50 TOPs)
- Ram: min. 32 GB, Disk: 50 GB
- OS: Ubuntu 22.04+
- Kernel with GTT support (e.g. v6.8)
- Connect your laptop to a charger

Costs (barebones, laptops)

- Ryzen 7 PRO 7840U (max. 65W): ~1400€ (03/2024)
- Ryzen AI 9 HX 370 (max. 135W): ~1400€
- Ryzen Al Max+ 395 (max. 140 – 320W): ~ 2300-3500€
- in comparison: Apple Mac Studio M4 Max 128 GB Ram: 4000-4500€

Integrated NPUs and GTT

- Unified memory: memory shared between CPU and GPU/NPU
- Graphics translation table (GTT): allows the graphics card direct memory access (DMA) to the host system memory



Setup Radeontop, Kernel

- Install radeontop
  - apt-get install radeontop
- Configure GTT in /etc/default/grub
  - GRUB_CMDLINE_LINUX_DEFAULT="amd_iommu=off ttm.pages_limit=6291456 ttm.pages_limit=6291456" (24 GB VRAM represented in KiB divided by 4)
- Update grub: sudo update-grub2
- Reboot

Check setup

Run radeontop, check GTT size:

```
                Graphics pipe   0,83%
                 Event Engine   0,00%

  Vertex Grouper + Tesselator   0,00%

            Texture Addresser   0,00%
                Texture Cache   0,00%

                Shader Export   0,00%
 Sequencer Instruction Cache   0,00%
          Shader Interpolator   0,00%
       Shader Memory Exchange   0,00%

               Scan Converter   0,00%
           Primitive Assembly   0,00%

                  Depth Block   0,00%
                  Color Block   0,00%
               Clip Rectangle   2,50%

            237M / 926M VRAM  25,59%
             49M / 24563M GTT   0,20%
  0,75G / 0,80G Memory Clock  93,33%
  0,80G / 2,70G Shader Clock  29,63%
```

Llama.cpp

- Open source project that runs inference on LLMs.
- Supports OpenAI-compatible endpoints like /v1/chat/completions
- Supports many hardware targets: e.g. x86, Metal, Cuda, Rocm, CANN, OpenCL, **Vulkan**
- Vulkan: cross-platform API and open standard for 3D graphics and computing.
- GGUF: file format is a binary format that stores both tensors and metadata in a single file

Setup Llama.cpp

- Download llama.cpp with Vulkan support
  - e.g. llama-b6585-bin-ubuntu-vulkan-x64.zip
- Extract the archive
  - e.g. unzip llama-b6585-bin-ubuntu-vulkan-x64.zip
- Start llama-cpp server
  - ./llama-server hf <modelname> <params>
- Open your browser with: **http://127.0.0.1:8080**

Start Llama.cpp with GPT-OSS

- Start llama-cpp server and download GPT-OSS 20b
  - ./llama-server -hf unsloth/**gpt-oss-20b-GGUF:F16** \
    --jinja -ngl 99 --threads -1 --parallel **4** --ctx-size **16384** \
    --temp 1.0 --top-p 1.0 --top-k 0 --no-mmap
    --kv-unified --n_predict **4096** \
    --chat-template-kwargs '{"reasoning_effort": "**low**"}'
- Open your browser with: **http://127.0.0.1:8080**

Start Llama.cpp with Qwen3

- Start llama-cpp server and download Qwen3
  - ./llama-server -hf unsloth/**Qwen3-4B-GGUF:UD-Q4_K_XL** \
    --jinja -ngl 99 --threads -1 --parallel **4** --ctx-size **262144** \
    --temp 0.7 --top-p 0.8 --top-k 20 --presence-penalty 1.0 --no-mmap \
    --kv-unified --cache-type-k q4_0 --cache-type-v q4_0 \
    --n_predict **4096**
- Open your browser with: **http://127.0.0.1:8080**
- Add „**/nothink**“ to your prompt to disable thinking

Testing GPT-OSS

Curl:

time curl -s http://127.0.0.1:8080/v1/chat/completions \
-H "Content-Type: application/json" -H "Authorization: Bearer no-key" \
-d '{
"model": "**unsloth_gpt-oss-20b-GGUF_gpt-oss-20b-F16.gguf**", "stream": **false**,
"messages": [{
    "role": "user", "content": "**When was Beethoven born?**"
}]}' | jq .

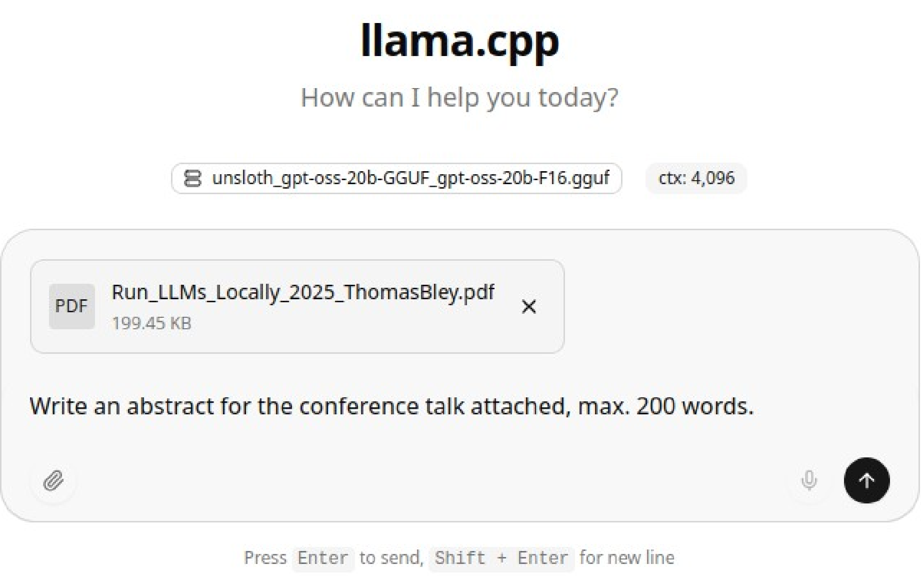

```
{
  "choices": [
    {
      "finish_reason": "stop",
      "index": 0,
      "message": {
        "role": "assistant",
        "reasoning_content": "Need answer: 1770-12-16.",
        "content": "Beethoven was born on **16 December 1770**."
```

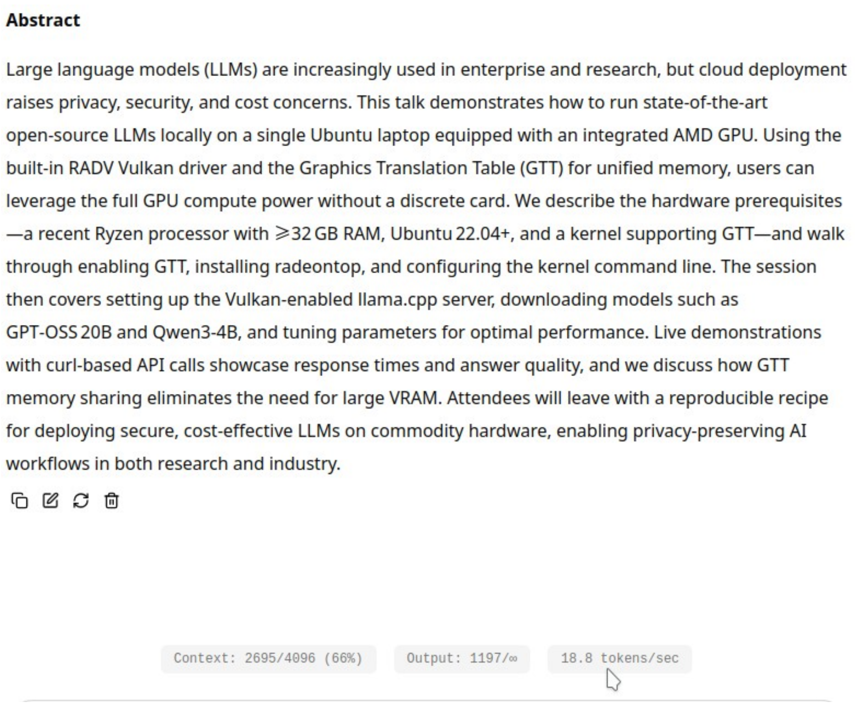
Testing Qwen3

Curl:

time curl -s http://127.0.0.1:8080/v1/chat/completions \
-H "Content-Type: application/json" -H "Authorization: Bearer no-key" \
-d '{
"model": "**unsloth_Qwen3-4B-GGUF_Qwen3-4B-UD-Q4_K_XL.gguf**", "stream": **false**,
"messages": [{
    "role": "user", "content": "**When was Beethoven born? /nothink**"
}]}' | jq .


```
{
  "choices": [
    {
      "finish_reason": "stop",
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "Ludwig van Beethoven was born on **December 16, 1770**."
```


llama.cpp
How can I help you today?
unsloth_gpt-oss-20b-GGUF_gpt-oss-20b-F16.gguf
ctx: 4,096
PDF
Run_LLMs_Locally_2025_ThomasBley.pdf
199.45 KB
Write an abstract for the conference talk attached, max. 200 words.
Press Enter to send, Shift + Enter for new line


Abstract
Large language models (LLMs) are increasingly used in enterprise and research, but cloud deployment raises privacy, security, and cost concerns. This talk demonstrates how to run state-of-the-art open-source LLMs locally on a single Ubuntu laptop equipped with an integrated AMD GPU. Using the built-in RADV Vulkan driver and the Graphics Translation Table (GTT) for unified memory, users can leverage the full GPU compute power without a discrete card. We describe the hardware prerequisites—a recent Ryzen processor with ⩾32 GB RAM, Ubuntu 22.04+, and a kernel supporting GTT—and walk through enabling GTT, installing radeontop, and configuring the kernel command line. The session then covers setting up the Vulkan-enabled llama.cpp server, downloading models such as GPT-OSS 20B and Qwen3-4B, and tuning parameters for optimal performance. Live demonstrations with curl-based API calls showcase response times and answer quality, and we discuss how GTT memory sharing eliminates the need for large VRAM. Attendees will leave with a reproducible recipe for deploying secure, cost-effective LLMs on commodity hardware, enabling privacy-preserving AI workflows in both research and industry.
Context: 2695/4096 (66%)
Output: 1197/∞
18.8 tokens/sec

Resources

- AMD Strix Halo Llama.cpp Toolboxes
- GLM 4.5-Air-106B and Qwen3-235B on AMD "Strix Halo" AI Ryzen MAX+ 395
- AMD Ryzen AI Max 395: GTT Memory Step-by-Step Instructions
- Wikipedia: Graphics Translation Table (GTT)
- unsloth: gpt-oss: How to Run & Fine-tune
- unsloth: Qwen3: How to Run & Fine-tune
- ViceVoice Text-to-Speech on Framework Desktop with Strix Halo

Thank You for listening!

Questions?

Slides: github.com/thomas-0816/talks/

Mastodon: phpc.social/@thbley